energywave.ai • Energy at the Point of Use — Version 2.1 (supersedes v2.0, August 2026)
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CONFIDENTIAL — FORECAST — FOR DISCUSSION · PREPARED FOR DEPARTMENT OF WAR TECHNICAL REVIEW

Energy at the point of use. The power bottleneck in ambulatory robotics, and a structured research program to address it.

Issuing entity: EnergyWave, Inc. (Delaware; formation in progress with K&L Gates LLP) — until formation, McLean Capital Development LLC (Wyoming), the xTech Search 10 applicant of record. Contact: Ian McLean, ian@energywave.ai, +1 (949) 280-0936. Version 2.1 supersedes v2.0 (August 2026); the change log is at the end.

> **Statement of claims discipline.** This document makes no performance claims about any device. EnergyWave has no prototype, no measured output and no validated energy-conversion mechanism, and nothing here should be read as an assertion that such a mechanism exists or has been demonstrated. What it asserts is narrower: (1) portable energy density is the binding constraint on ambulatory robotic platforms, and the constraint is worsening; (2) incremental electrochemical improvement will not close the gap on the timeline on which autonomy is arriving; (3) a small, disciplined research program with public retirement criteria is a low-cost, high-information way to test whether any non-chemical approach merits further national attention — and to close the question publicly if it does not. The field has a long history of unfalsifiable claims; this program intends to be legible as the opposite.

1 · THE BOTTLENECK

Progress in information technology has compounded for five decades. Progress in portable energy density has not. Transistor counts have risen by roughly seven orders of magnitude since 1971; practical specific energy in portable storage has risen by less than one. Every autonomous physical system inherits that divergence: the intelligence is arriving, the power to move it is not.

1.1 Current platform reality. Figures below are manufacturer or secondary reported statements as of the date shown, labelled per row at their provenance; none has been measured by EnergyWave.

Platform	Stated pack	Stated runtime	Source (date)
Tesla Optimus	2.3 kWh, 52 V	“about a full day of work”	Tesla AI Day via IEEE Spectrum (Sep 2022)
Figure 03	2.3 kWh	5 h at peak performance (≈ 460 W average)	Figure AI, “F.03 Battery Development” (Jul 2025)
Boston Dynamics Atlas (electric)	not stated	4 h; 2 h with heavy lifting; 1.5 h charge	Atlas spec sheet (Dec 2025)
Agility Digit	not stated	up to 4 h; deployed cycles ≈ 90 min run / 9 min charge	Agility (Mar 2025); IEEE Spectrum (Sep 2025)
Apptronik Apollo	not stated	4 h, swappable	The Robot Report (Aug 2023)
Unitree G1 · H1	≈ 0.43 kWh · 0.864 kWh	≈ 2 h (reseller) · not stated	unitree.com; RobotShop

The peer-reviewed literature on robot energetics puts humanoids at 150–400 W at rest against 60–80 W for a human, and 5–10× less efficient than humans in locomotion [12]. The gap between specification and field performance is itself a recognized problem: published runtimes are derived in controlled environments against favourable duty cycles. A fair summary is that today’s humanoids carry a few kilowatt-hours and run for hours, not

days, at a few hundred watts average; the industry's current answer to endurance is logistics — swap packs, dock, run in shifts — not energy.

1.2 Why actuation is the hard case. The load is not computation; it is mechanical work. Degrees of freedom added for dexterity are potentially major, task-dependent loads: holding a position under gravity draws power unless the actuator or transmission cancels or locks the load, and walking draws substantially more. Secondary coverage puts actuators at a large share of a production humanoid's bill of materials — a cost share, not a power share primary source to be confirmed. Moore's Law made the thinking cheap. It did nothing for the lifting.

1.3 Why chemistry will not close it. Lithium-ion is approaching its material ceiling: commercial cells were about 220 Wh/kg in 2020 against a program goal of 500 Wh/kg [13]; the best production cells reach 260–271 Wh/kg and packs 120–200 Wh/kg once housing, cooling and electronics are added [14]. Industry planning assumptions put improvement at roughly 4 % per year [6]; at that rate a doubling takes about eighteen years. Platform manufacturers are not waiting — they are engineering around it with inductive charging, swappable packs and high-nickel chemistries accepted for their density at a cost and thermal penalty. These are competent responses to a hard constraint, and an admission of it.

2 · WHY THIS IS A DEFENSE PROBLEM

The commercial case tolerates a charging dock; the defense case often does not. The dismounted power burden is a decades-old, unsolved logistics problem: a dismounted soldier in Afghanistan carried roughly 9.7 pounds of batteries, unable to recharge away from a vehicle or forward operating base, with battery weight projected to exceed 14 pounds for a 72-hour mission as every soldier is brought into the network [8]. Allied assessments place the burden as high as 25 % of a soldier's carried load on some mission profiles [9]. The Army has stated the trend plainly: across all six modernization priorities there has been a dramatic increase in demand for power as a result of energy-hungry new capabilities [10]. Add an ambulatory robotic platform to that formation and, at current energy density, the unit either tethers to a resupply and recharge cycle or does not deploy. Endurance becomes a doctrinal constraint, not an engineering footnote.

The strategic question follows: a five-fold improvement in portable energy density at the point of use changes force structure; a 4 %-per-year improvement does not. That gap is worth a small, well-governed research investment even at low probability of success, because the payoff is asymmetric and the cost of being second is high.

3 · WHAT IS PROPOSED

A research program, not a product program. **3.1 The protocol.** Every candidate mechanism proceeds through three gates, in order, with none skipped. *Characterize:* establish precisely what the claimed effect is, under what conditions, with what instrumentation, and what the null hypothesis predicts; reproduce the original measurement setup where records permit. *Conservation-test:* subject the effect to closed-system energy accounting with every input path measured — ambient thermal, RF, magnetic and mechanical included; if the output is accounted for by a conventional input, the mechanism is reclassified as a transducer and evaluated on those honest terms. *Extract or retire publicly:* an effect that survives proceeds to engineering characterization; one that does not is published as a negative result under the company's own name. Negative results in this field are almost never published, which is why the same claims recirculate for decades and why serious institutions will not engage. A retired claim is a deliverable.

3.2 The candidate threads. Three threads run under the one protocol: the founder's solid-state module in the Sweet lineage (thread A); an independent, pre-registered replication of a peer-reviewed report of unexplained output from a metal–insulator–metal diode under a Casimir cavity — 1.4 pW maximum inferred load power, one laboratory, not independently replicated in the reviewed evidence, with a published critique predicting the opposite sign (thread B); and the measurement platform itself, whose error budget is published as a methods paper before any result is claimed (thread C). The xTech Search 10 entry proposes thread B as an independent evaluation capability for Army sustainment, and that proposal stands as submitted. The founder's solid-state module (thread A) is the lead research

path — the founder's decision of 15 October 2026; a selected lead path is not measured performance or an operational schedule, and threads B and C remain distinct workstreams under the same protocol.

3.3 What is not claimed. The physics literature bearing on vacuum-energy extraction is not favourable: peer-reviewed reviews state that continuous cyclic extraction of this kind would violate the second law of thermodynamics, and that the kilowatt-scale extraction figures in that literature are unbuilt theoretical estimates rather than measured results. Moddel et al. report small electrical observations — picowatt-scale inferred load power — whose interpretation is disputed and whose independent net-cycle validation is not established in the reviewed evidence; those observations are distinct from any proof of a vacuum source. The company's theoretical physicist and its senior engineering advisor have been consistent on this point. Its theoretical physicist has assessed at least one historically prominent solid-state approach as non-viable as described; its senior engineering advisor, whose career runs from MIT research through commercialization of cryogenic systems for MRI and superconducting electronics, has held that the second law is not incomplete. The Department is not asked to accept a claim. It is asked whether competent instrumentation and honest accounting applied to specific historical claims, with publication in either direction, is worth doing.

4 · CONTROL AND SAFETY ARCHITECTURE

If any high-density portable energy source becomes practical, its control architecture is the enabling condition for release, and the program designs to that assumption from day one. **Platform binding:** the module is scoped to ambulatory robotic platforms and cryptographically bound to a specific host; it authenticates to the platform controller and delivers no output to an unrecognized host — a keyed consumable. **Application scoping in hardware:** power ceiling, ramp rate and duty envelope are fixed to the actuation and compute envelope of an ambulatory platform; repurposing as a bulk supply would require defeating hardware limits, a detectable and destructive act. **Tamper-responsive inertion:** envelope breach, unauthorized disassembly, out-of-spec thermal or electrical conditions or authentication failure trigger an irreversible transition to an inert state; no service mode, no recovery path, no field-repairable configuration. **Custody and revocation:** units are individually identified, attested and remotely revocable; a recovered or diverted unit is a dead unit.

Negative scope, intended as binding conditions of any government relationship: no directed-energy application; no propulsion application; no munitions or ordnance integration; no general-purpose or grid-scale configuration; no export or foreign transfer absent explicit U.S. Government authorization. The company would welcome these being written into an agreement. **Disclosure posture:** controlled rather than open; the time to establish government visibility is before results exist, not after. The Compartmentalization and Anti-Tamper Posture memo (v0.1, Sept 2026) sets out the bins, the kill, the platform binding and the questions for counsel.

5 · PROGRAM STRUCTURE

Phase	Content	Gate
Phase 0 — formation and instrumentation (0–6 months)	Delaware C-corporation formation (in progress with K&L Gates LLP); IP position through counsel; instrumentation specified and calibrated to conservation-test standard; independent measurement partner identified; advisory board seated.	Instrumentation validated against known standards before any candidate mechanism is energized.
Phase 1 — characterization (6–18 months)	Candidate mechanisms characterized and conservation-tested in priority order; each exits with a published determination.	At least one mechanism survives with independent witness, or comprehensive negative findings are published and the venture winds down or repurposes.

Phase	Content	Gate
Phase 2 — engineering characterization (18–36 months)	Contingent on Phase 1: scale, stability, thermal behaviour, manufacturability; the control architecture prototyped and adversarially tested independent of any energy result.	Reproducible bench performance under third-party observation.
Phase 3 — integration (36+ months)	Contingent on Phase 2: platform integration under a Government safety and security framework.	—

The control and safety architecture in §4 is developed on its own track and does not depend on Phase 1 succeeding; it has standalone value for any high-density portable source, including conventional ones.

6 · TEAM

Person	Line
Ian McLean — Founder	Wolfram Research; venture technology at Avnet with early engagements including Anduril and Wrap Technologies (NASDAQ: WRAP); FDIC during the 2008 financial crisis; founder of Statpedia; prior NDA consulting engagement with a U.S. intelligence organization.
Chip Story — Co-Founder, Physicist	Director, Zero Point Energy Project Group, a 501(c)(3) research organization; B.S. geological engineering (South Dakota School of Mines); M.S. geophysics (Colorado School of Mines); over fifty years in the oil and gas industry; published researcher.
Jack Sarfatti, Ph.D. — Team member, theoretical physicist	Theoretical physicist; B.A. physics, Cornell, trained under Hans Bethe; Ph.D. physics, UC Riverside; research fellow at Birkbeck College London with David Bohm and ICTP Trieste with Abdus Salam; leading member of the Fundamental Fysiks Group.
Alan Crunkleton, Ph.D. — Technical and financial advisor	Ph.D. Mechanical Engineering, MIT (1987); co-inventor of the Boreas cycle; co-founded Boreas, Inc. (1988), president; R&D 100 award (1994) for the B100 cryocooler.
Mike Wentz — Team member	Role to be announced.
Legal — K&L Gates LLP	Brian Novak, Ph.D., Partner in Charge, Irvine; Dennis Majewski, Partner, advanced degree in electrical and computer engineering, former senior electrical design engineer (semiconductors, quantum circuits, micro sensors). Handling the formation and the IP strategy.

Planned from the Seed Round. Arath Acosta: Design Engineer — a planned role, not a current employee.

7 · THE ASK

A technical review meeting to discuss: (1) whether the ambulatory-platform power constraint in §§1–2 is recognized as a capability gap within relevant program offices; (2) whether a conservation-tested characterization program with public retirement criteria is of interest as a low-cost, high-information investment; (3) whether the control architecture in §4 should be developed under Government specification, independent of the energy research outcome; (4) what framework would govern disclosure, custody and scope restriction should Phase 1 produce a positive result. The company is not asking for validation of a claim. It is asking whether this question is worth answering carefully, by people willing to publish a negative.

SOURCES

[6] Energy-density improvement projections, U.S. Patent 11542032 (hybrid-electric aircraft energy allocation) — secondary. [8] Meredith, “Soldier Power: A Growing Operational Concern,” Infantry Magazine, U.S. Army, Jul–Sep 2013. [9] “Portable power storage for the dismounted soldier,” European Security & Defence, Feb 2024. [10] “Army researchers hope to lighten Soldiers’ battery load,” U.S. Army, Aug 2018. [11] National Research Council, Energy-Efficient Technologies for the Dismounted Soldier; Meeting the Energy Needs of Future Warriors. [12] Kashiri et al., “An overview on principles for energy efficient robot locomotion,” Front. Robot. AI 5, 129 (2018). [13] U.S. DOE Battery500 progress update (May 2020). [14] BatteryDesign.net cell and pack benchmarks (2022–2026). Platform figures: Tesla AI Day via IEEE Spectrum (2022); Figure AI (2024, 2025); Boston Dynamics Atlas spec sheet (Dec 2025); Agility Robotics (2025); IEEE Spectrum (2025); The Robot Report (2023); unitree.com. v2.0’s sources [1]–[4] and [7] (optimusk.blog, theresarobotforthat.com, RobotWale, nexthumanoid.com) are replaced by the primary and peer-

reviewed sources above. This document contains no classified information and no export-controlled technical data; it describes no device design, fabrication method or operating parameter.

CHANGE LOG — V2.0 (AUGUST 2026) TO V2.1

“Department of Defense” → “Department of War” in narrative; agency and form names unchanged where quoted.

Issuing entity: EnergyWave, Inc. on formation; until then the xTech applicant of record; contact ian@energywave.ai; v2.0’s letterhead removed.

§1.1 platform figures replaced with dated manufacturer statements; blog sources dropped; §1.2 actuator figures qualified as a cost share.

§3.2 added: the three research threads under one protocol; the lead path is the founder’s solid-state module (15 October 2026); the xTech proposal of thread B stands as submitted.

§3.3 “principal theoretical physicist” → “physics advisor”, matching the October deck.

§6 team updated to the October 2026 roster.

§7 item numbering corrected.

Cross-reference to the Compartmentalization and Anti-Tamper Posture memo added in §4.

Basis: the August 2026 white paper v2.0 (full text); the Technical Prospectus v1 §3 and Appendix A; the 13 October 2026 deck, slide 12; the Team and Counsel page; the Compartmentalization and Anti-Tamper Posture memo; the Zero Point Energy Project Group’s published pages; the decisions of 15–16 October 2026.